

YESI-FULFILLMENT

Vortex WMS Master Catalog

Complete Workflows, Functions & Tips

Ozon Fulfillment | Google Sheets | Inventory | Multi-Device

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Project: warehouse_ops_pro/app

Stack: React 19 + Vite + Dexie + PWA + Tauri

Table of Contents

1. Quick Start & Installation
2. Login, Roles & Security
3. Navigation & Keyboard Shortcuts
4. Dashboard & Notifications
5. Ozon Google Sheets Pipeline (Complete)
6. Inbound Receiving Workflow
7. Putaway Management
8. Pick & Pack Workflow
9. Batch Pick Center
10. Inventory Hub
11. Serial Tracking & IMEI
12. Shipping & Carrier Rates
13. Posting Tracker (Ozon Proof)
14. PDF Sequencer & Label Tools
15. Returns Processing
16. Cycle Count & Replenishment
17. QC Management
18. Dock & Yard Management
19. Labor Management
20. Analytics & KPIs
21. Integrations Hub
22. Guardian Ops & Pressure Monitor
23. Vortex AI Assistant
24. Manual Order Tools
25. Administration
26. Database Schema (All 34 Tables)
27. Library Functions Reference
28. Automation & Multi-Device
29. PWA, Offline & Desktop (Tauri)
30. Tips, Tricks & Troubleshooting
31. Daily Warehouse Playbooks

1. Quick Start & Installation

Prerequisites

- Node.js 18+ and npm
- Modern browser (Chrome, Edge, Firefox) or Windows for Tauri desktop

Install & Run (Web)

```
cd warehouse_ops_pro/app
npm install
npm run dev          # http://localhost:3000
npm run build        # Production PWA build
npm run preview      # http://localhost:4173
```

Desktop App (Tauri)

```
npm run tauri:dev    # Native Windows shell
npm run tauri:build  # .exe / .msi installer
```

First Login

- Admin: VINOVJ / VINOVJ (full access)
- Operator: operator / operator
- Supervisor: supervisor / supervisor

On first launch, seedDatabase() populates inventory, orders, zones, Ozon sheet config, and integration endpoints automatically.

2. Login, Roles & Security

Roles

Role	Access
admin	All pages + User Mgmt, SIM, Guardian, Ad
supervisor	All operational pages, no admin section
operator	Standard warehouse tasks, no admin secti

Session Security

- 30-minute inactivity timeout (auto logout)
- Session stored in localStorage key: vortex_session
- All login/logout events logged to auditLogs table
- Admin-only: Google API keys, auto-sync config, user CRUD

3. Navigation & Keyboard Shortcuts

Global Shortcuts

Shortcut	Action
Ctrl/Cmd + K	Open Command Palette (fuzzy search all p
Escape	Close Command Palette / modals
Up / Down	Navigate palette items
Enter	Select palette item / submit barcode sca

Sidebar Structure (4 Tiers)

- Operations: Dashboard, Inbound, Putaway, Pick&Pack, Batch Pick, Returns, Shipping, Dock, Posting, Ozon Sheets
- Advanced: Inventory, Serial, Cycle Count, Replenishment, Labor, Analytics, QC, Integrations, Memory, Scanner
- Manual Orders: PDF Sequencer, Templates, Bulk Converter, Auditor
- Administration (admin only): Users, SIM Manager, Guardian, Vortex AI, Barcode Generator

4. Dashboard & Notifications

Route: / | Real-time KPIs with auto-refresh (configurable in Settings).

KPI Cards

- Total stock, low-stock count, pending orders, shipped today
- Returns, QC holds, cycle count pending, wave batches active
- Posting count, worker tasks, fulfillment rate

Notification Bell (TopBar)

- Unread count badge; click to open alert panel
- Sources: Ozon sheet sync, low stock, high order volume, Guardian pressure
- Types: success, warning, error, info - persisted in Zustand store

Sync Status Indicator

- ONLINE - SYNCED (green) when connected
- OFFLINE - LOCAL MODE (red banner) - all data still works via IndexedDB

5. Ozon Google Sheets Pipeline (Complete)

Route: /ozon-sheets | The core Ozon fulfillment integration.

Pre-Configured Sheet

ID: 1NNSAHZ8A7l2nDbwZ-9tLE6K5lmdWo7V6MBizbhH875o
GID: 1858822101

Pipeline Flow

Google Sheet -> Column Mapper -> WMS Orders -> Inventory -> Pick & Pack -> Posting Tracker

Three Sync Methods

- LIVE URL (gviz CSV): Sheet must be published 'Anyone with link -> Viewer'
- API KEY: Google Sheets API v4 - works with private sheets (admin sets key)
- CSV UPLOAD: File -> Download -> CSV from Google Sheets - always works offline

Column Auto-Mapping (RU/EN)

- orderId: order id, posting, otpravlenie, shipment
- sku: sku, artikul, article, kod tovara
- product: product, tovar, naimenovanie
- quantity: qty, kolichestvo, count
- status: status, status (RU), fulfillment
- tracking: tracking, trek, ozon id
- city: city, gorod, sklad

Sync Behavior (runSheetSync)

- Upserts orders into db.orders (import new, update existing)
- Creates inventory stubs for unknown SKUs at OZON-PENDING location
- Dedup via sheetSyncRows.rowHash when skipDuplicates enabled
- Optional autoAllocatePendingOrders() for FEFO/FIFO reservations
- Logs every sync to sheetSyncLogs with device ID and duration

Automation (ozonAutomation.ts)

- Runs every 5 min from Layout on login
- Auto-syncs sheets where autoSync=1 and interval elapsed
- Low-stock alerts when stock <= reorderPoint
- High volume alert when pending orders > 20
- Guardian critical pressure notifications

Admin Settings

- Sheet URL, fetch method (auto/gviz/api), sync interval (5-120 min)
- Toggles: auto-sync, auto-allocate, skip duplicates

- Google API key (password field, admin only)

Multi-Device

- Each browser gets unique device ID (localStorage vortex_device_id)
- Heartbeat every 60s registers operator + platform in connectedDevices
- Online = last seen within 5 minutes

6. Inbound Receiving Workflow

Route: /inbound | Three receiving modes.

Manual Receive

- Scan/enter SKU, quantity, bin location, lot number, expiry date
- Optional cross-dock order ID for direct-to-ship
- QC mode: pending / passed / failed
- Creates inbound record + updates inventory + inventoryMovement

Excel Upload

- Upload .xlsx, map columns, preview rows, bulk commit
- Uses xlsx library for parsing

PO Reconciliation

- Match received qty against purchaseOrders
- Variance tracking, supplier info, putaway suggestions

G-Sheets Link Rack

- Bookmark Google Sheet URLs in localStorage for quick access

7. Putaway Management

Route: /putaway | Directed putaway task queue.

Task Lifecycle

pending -> assigned -> in_progress -> completed (or cancelled)

Source Types

- receiving, returns, transfer, qc_release, cross_dock

Putaway Methods

- direct: SKU to specific bin
- zone: zone-level placement
- cross_dock: bypass storage to outbound
- consolidation: merge into existing location

Performance Tracking

- Estimated vs actual time, distance walked, QC status per task

8. Pick & Pack Workflow

Route: /pick-pack | Core order fulfillment.

Standard Pick

- 1. Select pending order from queue (sorted by priority)
- 2. Scan each required SKU (barcode wedge or manual)
- 3. System decrements inventory, logs pick movement
- 4. Mark order Shipped when all SKUs confirmed

Wave Pick Mode

- Aggregate SKUs across multiple orders
- Single pass through warehouse for shared items

Order Statuses

Pending, Picking, Packed, ReadyToShip, Shipped, Returned, Cancelled, QCHold, CrossDock

Priority Levels

- urgent (red) - express Ozon orders
- high (yellow) - picking in progress or high priority sheet flag
- normal (default)

9. Batch Pick Center

Route: /batch-pick | Advanced multi-order picking.

Workflow

- 1. Select multiple pending orders
- 2. optimizePickPath() - nearest-neighbor TSP route
- 3. createBatchGroup() - batch ID with pick path JSON
- 4. Optional autoAllocatePendingOrders() for FEFO reservations

Pick Path Optimizer

- Calculates total distance, estimated time, zone profile
- Stores route in pickPaths and batchGroups tables

10. Inventory Hub

Route: /inventory | Full stock management.

Features

- CRUD: add/edit/delete SKUs with full metadata
- ABC classification, velocity (high/medium/low)
- Lot/batch/expiry tracking, FEFO priority
- Barcode, RFID tag, dimensions, cost per unit
- Reorder point and reorder quantity alerts
- Movement history viewer
- Zone capacity map visualization

Stock Adjustment

- Manual stock edit with audit trail via inventoryMovements

11. Serial Tracking & IMEI

Route: /serial-tracking | Full traceability for serialized items.

Serial Statuses

in_stock, reserved, shipped, returned, defective, quarantine

Phone-Specific Fields

- IMEI1, IMEI2, TAC prefix, color, storage, warranty dates

SIM Manager (/sim-manager)

- TAC prefix database for model identification
- Luhn algorithm IMEI calculator
- Batch IMEI generation with correct check digits

12. Shipping & Carrier Rates

Route: /shipping | Rate shopping and label generation.

Carriers (Seeded)

- FedEx (Ground, 2Day), UPS (Ground, Next Day), USPS Priority, DHL Express, Ozon Standard

Box Sizing

- 6 box sizes from padded envelope to extra large
- Auto-suggest based on order weight/dimensions

Label Workflow

- Generate shippingLabels record, print modal, mock tracking number

13. Posting Tracker (Ozon Proof)

Route: /posting-tracker | Ozon fulfillment proof-of-posting.

Capture Workflow

- 1. Enter posting ID and tracking ID
- 2. Select status: received, in_transit, delivered, posted, exception
- 3. Capture photo and/or video proof
- 4. Auto-capture geolocation (city, coordinates)
- 5. Organize by folder (e.g. 'June 2024')

Export

- ZIP export: JSON metadata + photo/video files per posting
- Search and filter by status, folder, posting ID

14. PDF Sequencer & Label Tools

PDF Sequencer (/pdf-sequencer)

- Upload Ozon label PDF, extract tracking IDs via pdfjs
- Paste master sequence list, reorder pages to match
- Export sorted PDF for printing in pick order

Barcode Generator (/barcode-generator)

- Location, product, serial, multi-item, batch label generation
- JSBarcode + QRCode output, print-ready layouts

Barcode Scanner (/barcode-scanner)

- Camera or file scan -> lookup inventory/orders

15. Returns Processing

Route: /returns

- Scan returned SKU, select reason code
- Restock to inventory OR send to QC hold if defective
- Updates order status to Returned
- Creates return record + inventory movement

16. Cycle Count & Replenishment

Cycle Count (/cycle-count)

- Tabs: List, Create, By Location, Dashboard
- ABC-classified scheduling (A=weekly, B=monthly, C=quarterly)
- Variance detection and supervisor approval

Replenishment (/replenishment)

- Low-stock alerts from reorder points
- Replenishment task creation and slotting suggestions
- Zone capacity utilization checks

17. QC Management

Route: /qc | Quality control holds and releases.

- Inbound QC queue: pending items from receiving
- QC holds: block SKUs from picking until released
- Release to inventory or reject to defective status
- Links to orders on QCHold status

18. Dock & Yard Management

Route: /dock | Inbound appointment scheduling.

Appointment Lifecycle

scheduled -> checked_in -> unloading -> completed (or no_show / cancelled)

Fields

- Supplier, PO number, dock number, carrier, trailer, pallet count, ASN

19. Labor Management

Route: /labor | Worker performance and task assignment.

- UPH (units per hour), accuracy %, distance walked
- Task types: pick, pack, receive, putaway counts
- 7-day performance history with leaderboards
- Worker skill levels (1-5) and zone assignments

20. Analytics & KPIs

Route: /analytics | Charts and reports via Recharts.

- Fulfillment rate trends
- Inventory value by category
- Movement volume over time
- Posting status distribution
- Wave batch completion rates

21. Integrations Hub

Route: /integrations | External system connectors.

Seeded Endpoints

- SAP ERP, Shopify OMS, FedEx TMS, Amazon Marketplace, Ozon Marketplace

Ozon Real Sync

- Click Sync on Ozon Marketplace -> triggers runSheetSync() with real Google Sheet import
- Other endpoints use simulated sync wizard (for demo)

Endpoint Management

- Add/edit/toggle endpoints, configure sync interval, view error counts

22. Guardian Ops & Pressure Monitor

Route: /guardian | Admin-only warehouse health monitor.

Pressure Scenarios (detectPressureScenarios)

- Stockout risk, SLA breach, worker overload, QC backlog, zone overcapacity
- Levels: normal, warning, critical, emergency

Adaptation Engine

- Modes: normal, efficiency, surge, crisis
- Auto-suggestions based on time-of-day and order volume

System Maintenance

- Retention-based cleanup of old audit logs, movements, tasks
- Storage usage monitor, configurable retention days

23. Vortex AI Assistant

Access: Sidebar 'Vortex AI', Command Palette, floating panel.

Supported Intents (Rule-Based, No API Key Needed)

- Stock lookup: 'where is APP-IP15-256-BLK'
- Low stock: 'low stock', 'reorder'
- Order status: 'ORD-9984', 'pending orders'
- Pressure: 'warehouse pressure', 'guardian'
- Putaway: 'putaway queue'
- Serial/IMEI: 'IMEI', 'SN-...'
- Batch pick: 'batch pick', 'wave'
- Dock: 'dock schedule'
- Allocation: 'allocate', 'FEFO'
- Labor: 'UPH', 'worker productivity'
- Navigate: 'go to shipping'
- Analytics: 'KPI', 'reports'

Suggested Prompts

- What is the warehouse pressure?
- Show low stock items
- Putaway queue status
- Optimize pick path
- Dock schedule today
- Order pipeline summary

24. Manual Order Tools

Templates (/templates)

- Raw product title -> standardized title mapping

Bulk Converter (/bulk-convert)

- Batch title standardization, multi-language translation
- Barcode/QR batch generation with ZIP export

Auditor (/auditor)

- Compare master list vs scanned list, highlight discrepancies

Memory (/memory)

- Aliases and preferences (site name, operator name)

25. Administration

User Management (/users)

- CRUD users with roles, view full audit log

Settings (/settings)

- Profile display name update
- Export all data as JSON (exportAllData)
- Import JSON backup
- Database statistics viewer
- Reset database (destructive)
- Auto-refresh interval, low stock threshold

26. Database Schema (All 34 Tables)

Database: VortexWMS (Dexie/IndexedDB) | Current Version: 8

Table	Key Fields
inventory	sku, location, stock, barcode, lot, expi
orders	orderId, status, priority, requiredSkus,
returns	orderId, sku, reason, restocked, quantit
inbound	sku, qty, bin, qcStatus, poNumber, suppl
auditLogs	action, details, operator, timestamp
users	username, password, role, displayName, s
inventoryMovements	sku, type, quantity, orderId, operator
cycleCounts	sku, location, status, abcClass
zoneCapacities	zone, maxCapacity, category, binType
workerTasks	type, status, assignedTo, orderId
replenishmentTasks	sku, status, suggestedQty
qcHolds	sku, status, reason, orderId
waveBatches	waveId, status, orderIds
templates	raw, standard product title mapping
aliases	source, target text aliases
preferences	key, value settings
simDb	tacPrefix, modelSeries, expectedOffset
postingRecords	postingId, trackingId, status, photoData
purchaseOrders	poNumber, supplier, status, items JSON
shippingLabels	orderId, carrier, trackingNumber, status
carrierRates	carrier, service, zone, rate, weightFrom
workerPerformance	workerId, date, uph, accuracy, distance
integrationEndpoints	name, type, provider, status, lastSync
batchGroups	batchId, orderIds, pickPath, pickingMeth
pickPaths	pathId, route JSON, totalDistance, picke
boxSizes	name, dimensions, maxWeight, material
serialNumbers	serialNumber, sku, imei1, status, warran
putawayTasks	taskId, sku, destinationZone, sourceType
dockAppointments	appointmentId, dockNumber, scheduledDate
inventoryReservations	reservationId, orderId, fefoPriority
sheetSyncConfigs	spreadsheetId, autoSync, syncInterval, a

sheetSyncLogs	configId, status, recordsProcessed, devi
sheetSyncRows	rowHash, orderId (dedup tracking)
connectedDevices	deviceId, operator, lastSeen, platform

27. Library Functions Reference

Module	Key Exports
db.ts	db, logAction, logInventoryMovement, exp
auth.tsx	AuthProvider, useAuth, login, logout, is
store.ts	useAppStore, addNotification, commandPal
allocation.ts	allocateOrder, autoAllocatePendingOrders
pickPathOptimizer.ts	optimizePickPath, createPickPath, create
criticalWorkflow.ts	detectPressureScenarios, calculatePressu
adaptation.ts	calculateAdaptation, getCurrentAdaptatio
maintenance.ts	runSystemCleanup, getStorageUsage, autoC
aiAssistant.ts	processUserMessage, SUGGESTED_PROMPTS
googleSheets.ts	fetchSheet, parseSheetUrl, parseUploaded
ozonSheetMapper.ts	detectColumnMapping, mapRowsToOzonOrders
ozonSheetSync.ts	runSheetSync, writeBackOrderStatus, getD
ozonAutomation.ts	runAutomationChecks, startAutomationLoop
deviceRegistry.ts	getDeviceId, registerDeviceHeartbeat, ge

28. Automation & Multi-Device

Background Automation (starts on login)

- Layout.tsx starts startAutomationLoop(operator, 5) - checks every 5 min
- Device heartbeat every 60 seconds
- Stops on logout / unmount

What Runs Automatically

- Sheet auto-sync (if enabled, per config interval)
- Low-stock notifications
- Guardian critical pressure alerts
- High pending order volume warnings (>20)

Multi-Device Coordination

- Each device: unique ID, platform detection (iPhone/Android/Windows/Mac)
- Ozon Sheets Hub shows all warehouse devices with online status
- Sync logs include deviceId for audit trail

29. PWA, Offline & Desktop (Tauri)

Offline-First Architecture

- ALL data in IndexedDB - fully functional without internet
- Sheet fetch and API calls require network; CSV upload works offline

PWA Features

- Service worker via vite-plugin-pwa (Workbox auto-update)
- Install prompt on supported browsers
- Manifest: YESI-FULFILLMENT, standalone, landscape-primary

Tauri Desktop

- Native Windows .exe and .msi installers
- Same web UI in embedded webview with native shell

Data Backup

- Settings -> Export JSON exports all 34 tables
- Import JSON to restore on new device/browser

30. Tips, Tricks & Troubleshooting

Ozon Sheet Sync Issues

- PROBLEM: 'Sheet is private' -> SOLUTION: Publish sheet OR use CSV upload OR add API key
- PROBLEM: Columns not detected -> SOLUTION: Check headers match RU/EN aliases; use Preview Columns
- PROBLEM: Duplicate orders -> SOLUTION: Enable skipDuplicates in admin config
- PROBLEM: No inventory for pick -> SOLUTION: Receive stock via Inbound OR check auto-allocate

Performance Tips

- Use Batch Pick for 5+ orders in same zone - saves 30-40% walk time
- Enable auto-allocate on sheet import for immediate pick readiness
- Set reorder points on high-velocity SKUs to get early low-stock alerts
- Run Guardian maintenance weekly to keep IndexedDB lean

Mobile / iPhone Tips

- Install as PWA for fullscreen warehouse use
- Posting Tracker camera requires HTTPS on mobile
- Barcode wedge scanners work via BarcodeReader Enter-to-submit

Power User Shortcuts

- Ctrl+K -> type 'ozon' -> jump to Ozon Sheets Hub instantly
- Ctrl+K -> type 'batch' -> Batch Pick Center
- Ctrl+K -> type 'pressure' -> Guardian Dashboard
- Vortex AI -> 'optimize pick path' for route suggestions

Data Management

- Export JSON before browser cache clear or DB reset
- Audit logs track every sync, login, and inventory change
- sheetSyncLogs provide per-device sync history for debugging

31. Daily Warehouse Playbooks

Morning Startup (Admin)

- 1. Login as VINOVI
- 2. Check Dashboard KPIs and notification bell
- 3. Open Ozon Sheets Hub -> Sync Now (or verify auto-sync ran)
- 4. Review Guardian pressure - resolve any critical alerts
- 5. Check Dock schedule for inbound appointments

Order Fulfillment (Operator)

- 1. Ozon Sheets sync brings new orders overnight
- 2. Batch Pick Center -> select pending orders -> optimize path
- 3. Pick & Pack -> scan SKUs -> mark Shipped
- 4. PDF Sequencer -> sort Ozon labels in pick order
- 5. Posting Tracker -> photo proof + geolocation per posting

Inbound Day

- 1. Dock -> check-in supplier appointment
- 2. Inbound Receiving -> scan pallets, PO reconcile
- 3. Putaway -> complete directed tasks to bins
- 4. QC -> release passed items, hold failed

End of Day

- 1. Cycle Count -> complete scheduled ABC counts
- 2. Analytics -> review fulfillment rate
- 3. Settings -> Export JSON backup
- 4. Guardian -> run system cleanup if needed

Private Sheet Workflow (Recommended)

- 1. In Google Sheets: File -> Download -> CSV
- 2. Ozon Sheets Hub -> CSV Upload tab -> drop file
- 3. Review column mapping preview
- 4. Orders appear in Pick & Pack within seconds